1) $y - x\frac{dy}{dx} = \left(\frac{dy}{dx}\right)^2$

$$\frac{dy}{dx} = p$$

$$y - xp = p^2$$

$$y = p^2 + xp$$

$$\frac{dy}{dx} = 2p\frac{dp}{dx} + p + x\frac{dp}{dx}$$

$$\cancel{p} = 2p\frac{dp}{dx} + \cancel{p} + x\frac{dp}{dx}$$

$$0 = \frac{dp}{dx} \rightarrow p = c$$

$$y - xc = c^2$$

$$y = xc + c^2$$

2) $y + \left(\frac{dy}{dx}\right)^2 = x\frac{dy}{dx}$

$$\frac{dy}{dx} = p$$

$$y + p^2 = xp$$

$$y = xp - p^2$$

$$\frac{dy}{dx} = x\frac{dp}{dx} + p - 2p\frac{dp}{dx}$$

$$0 = x\frac{dp}{dx} - 2p\frac{dp}{dx}$$

$$dp = 0$$

$$p = c$$

$$y = xc - c^2$$

3) $y = x\frac{dy}{dx} + \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$

$$\frac{dy}{dx} = p$$

$$y = xp + \sqrt{1 + p^2}$$

$$\frac{dy}{dx} = x\frac{dp}{dx} + p + \frac{1}{2}\frac{2p\frac{dp}{dx}}{\sqrt{1+p^2}}$$

$$\cancel{p} = x\frac{dp}{dx} + \cancel{p} + \frac{\frac{dp}{dx}}{\sqrt{1+p^2}}$$

$$0 = \frac{dp}{dx}\left(x + \frac{1}{\sqrt{1+p^2}}\right)$$

$$\frac{dp}{dx} = 0$$

$$p = c$$

$$y = xc + \sqrt{1+c^2}$$

4) $y = x\dfrac{dy}{dx} + \dfrac{dy}{dx}$ $\dfrac{dy}{dx} = p$

$$y = xp + p$$

$$\frac{dy}{dx} = x\frac{dp}{dx} + p + \frac{dp}{dx}$$

$$\cancel{p} = x\frac{dp}{dx} + \cancel{p} + \frac{dp}{dx}$$

$$0 = \frac{dp}{dx}(x+1)$$

$$\frac{dp}{dx} = 0$$

$$p = c$$

$$y = xc + c$$